Target-Specific Recoverability Maps for Reduced-Lead ECG Reconstruction

A graded per-lead identifiability + conditioning certificate with calibrated intervals, and an honest account of a retracted fabrication claim

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Abstract

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature, on *which* lead, a clinician can trust. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S, and a target lead ℓ, two closed-form numbers from one truncated SVD of the segment's empirical rank-3 spatial sub-matrix—a graded *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target's low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$. The per-lead criterion is the classical estimability condition specialized to ECG; the contribution is its use as a graded, *empirically-estimated* map (the estimated subspace differs 43–55° from the textbook inverse-Dower dipole), the distribution-free calibrated intervals we attach to it, and a reconstructor-*invariant* safety certificate. On PTB-XL the map ranks target leads on a continuum—from limb leads, anterior precordial ST (V1–V4) is unidentifiable while lateral V5/V6 are near-identifiable—and the total ST-threshold error floor is near-invariant across reconstructors (47.1–50.1%) while the false-positive/false-negative split is a design choice. The certificate transfers across PTB-XL and Chapman for QRS (aligned subspaces, reproduced κ ordering) but not in the ST/T third direction. Finally, we document in full a claim this project retracted—that a diffusion reconstructor proves "fabrication is a property of the objective"—which a leakage-corrected, multi-seed re-analysis reduced to a marginal ($\approx 1.3\sigma$) effect with no supporting realism trend. This long version accompanies a four-page conference paper and contains the full proofs, ablations, cross-dataset study, and the negative result.

1 Introduction

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads, and a reconstruction that gets the mean right but invents an ST deviation invents a finding.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead's low-rank morphology is even identifiable, and how well-conditioned is its recovery?* Each waveform segment's instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis $\boldsymbol{M}_s$ (top-3 singular vectors of the population lead covariance). We stress that $\boldsymbol{M}_s$ is a *population-estimated (PCA)* object: on real PTB-XL it shares two directions with the classical vectorcardiographic (inverse-Dower) dipole but its third direction differs materially, so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 5).

This document is the extended version of a four-page conference submission. It adds: the full statement and proof of the certificate (Sec. 3, App. A); the cross-dataset transfer study (Sec. 6); and a complete account of a claim we retracted during pre-submission review (Sec. 7). We report the retraction in detail on purpose: hiding the earlier, more exciting numbers would be precisely the selective reporting our certificate is designed to expose.

2 Related Work

Lead reconstruction and VCG synthesis. Fixed linear transforms synthesize the vectorcardiogram or missing leads from a standard set (inverse-Dower [3], Kors regression [4]), and reduced-lead reconstruction has a long clinical literature [5]; recent deep models push reconstruction accuracy [1, 2]. These target the reconstruction; we target the *prior* question of what a given configuration can support, independent of the reconstructor. **Identifiability and lead selection.** Whether a linear functional is recoverable from a sub-observation is the classical estimability question [6]; choosing which leads to observe is optimal sensor selection [7]. Our per-lead η is estimability specialized to the ECG dipolar functional; the novelty is not the identity but the graded, empirically-estimated, per-target-lead map with calibrated per-record error attached. **Uncertainty for inverse problems.** Conformal prediction [8, 9] gives distribution-free intervals under exchangeability; we apply Mondrian CQR per (S, s, ℓ) group.

3 The Recoverability Certificate

Model. Within segment s write the population 12-lead vector as $\boldsymbol{L}_s = \boldsymbol{\mu}_s + \boldsymbol{M}_s\boldsymbol{d} + \boldsymbol{r}_s$, where $\boldsymbol{M}_s \in \mathbb{R}^{12\times 3}$ is the (population-estimated, PCA) dipolar basis, $\boldsymbol{d}$ the dipole coordinates, and $\boldsymbol{r}_s$ the non-dipolar residual. Observing S gives $\boldsymbol{y}_S = \mathbf{Sel}_S\boldsymbol{L}_s + \boldsymbol{n}$. Let $\boldsymbol{M}_{s,S} = \mathbf{Sel}_S\boldsymbol{M}_s$ have truncated SVD at relative tolerance ϱ; $\boldsymbol{M}_{s,S}^+$ its truncated pseudo-inverse and $\boldsymbol{P}_{\text{obs}} = \boldsymbol{M}_{s,S}^+\boldsymbol{M}_{s,S}$ the projector onto the observed dipole coordinate directions.

Proposition 1 (Per-target-lead dipolar identifiability and conditioning)**.** *The mean-centred dipolar estimate $\widehat{\boldsymbol{L}}_s = \boldsymbol{\mu}_s + \boldsymbol{M}_s\boldsymbol{M}_{s,S}^+(\boldsymbol{y}_S - \boldsymbol{\mu}_{s,S})$ satisfies, for each target lead ℓ with $\boldsymbol{e}_\ell$ the ℓ-th unit vector:*

(i) Identifiability. *Define $\eta_{s,\ell}(S) = \left\|\boldsymbol{e}_\ell^\top\boldsymbol{M}_s(\boldsymbol{I} - \boldsymbol{P}_{\text{obs}})\right\|_2$. If $\eta_{s,\ell}(S) = 0$, the dipolar component of lead ℓ is identifiable from S: $\boldsymbol{e}_\ell^\top\boldsymbol{M}_s\boldsymbol{d}$ is determined by $\mathbf{Sel}_S\boldsymbol{M}_s\boldsymbol{d}$. If $\eta_{s,\ell}(S) > 0$, there is a dipole direction unobserved by S that changes lead ℓ; its dipolar component is not identifiable at any SNR.*

(ii) Conditioning. *On the identifiable part, the error from observation noise and from the observed non-dipolar residual is amplified into lead ℓ by $\kappa_{s,\ell}(S) = \left\|\boldsymbol{e}_\ell^\top\boldsymbol{M}_s\boldsymbol{M}_{s,S}^+\right\|_2$: $\left|\boldsymbol{e}_\ell^\top(\widehat{\boldsymbol{L}}_s - \boldsymbol{\mu}_s - \boldsymbol{M}_s\boldsymbol{P}_{\text{obs}}\boldsymbol{d})\right| \le \kappa_{s,\ell}(S)\left(\|\mathbf{Sel}_S\boldsymbol{r}_s\| + \|\boldsymbol{n}\|\right)$.*

The global $\kappa_s(S) = \|\boldsymbol{M}_s\boldsymbol{M}_{s,S}^+\|_2 = \max_\ell \kappa_{s,\ell}(S)$ is a configuration-level worst case. Both $\eta_{s,\ell}$ and $\kappa_{s,\ell}$ depend on the estimated $\boldsymbol{M}_s$ and on ϱ; near-rank-deficient S are reported with an ϱ-sensitivity sweep and record-bootstrap confidence intervals.

Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [6]) specialized to the per-lead linear functional $\boldsymbol{e}_\ell^\top\boldsymbol{M}_s\boldsymbol{d}$: $\eta_{s,\ell}(S) = 0$ iff that functional lies in the row space of $\boldsymbol{M}_{s,S}$, and $\kappa_{s,\ell}$ is its coefficient norm. We claim no novelty for this identity. The contribution (Sec. 5) is its use as a *graded*, per-target-lead, *empirically-estimated* recoverability map: a continuous η that ranks target leads (not a binary rank test), a subspace $\boldsymbol{M}_s$ that differs materially from the textbook inverse-Dower dipole, and per-record calibrated error rates (Sec. 4) attached to it—quantities not read off lead geometry.

The non-dipolar residual (no independence claim on real data). Let $\boldsymbol{m}_s(\boldsymbol{y}_S) = \mathbb{E}[\boldsymbol{r}_s \mid \boldsymbol{y}_S]$ and $\boldsymbol{u}_s = \boldsymbol{r}_s - \boldsymbol{m}_s(\boldsymbol{y}_S)$, so $\mathbb{E}[\boldsymbol{u}_s \mid \boldsymbol{y}_S] = \mathbf{0}$. We do *not* assume $\boldsymbol{u}_s \perp \boldsymbol{y}_S$. The Bayes-optimal reconstruction error of the residual is

$$\inf_{\hat{\boldsymbol{r}}} \mathbb{E}\big\|\boldsymbol{r}_s - \hat{\boldsymbol{r}}(\boldsymbol{y}_S)\big\|^2 = \mathbb{E}\big[\operatorname{Var}(\boldsymbol{r}_s \mid \boldsymbol{y}_S)\big].$$

Part of $\boldsymbol{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\boldsymbol{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—*not* declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only)**.** *If the data are generated so that a component $\boldsymbol{u}$ is statistically independent of $\boldsymbol{y}_S$ (an explicit synthetic construction), then $p(\boldsymbol{u} \mid \boldsymbol{y}_S) = p(\boldsymbol{u})$, the Bayes estimate of $\boldsymbol{u}$ is its prior mean, and $\mathbb{E}\|\widehat{\boldsymbol{u}} - \boldsymbol{u}\|^2 \geq \operatorname{Var}(\boldsymbol{u})$ for any estimator. This strict limit is invoked* only *for the synthetic model; on real ECG we make no such independence claim.*

4 Distribution-Free Calibration

4.1 Calibrated intervals for the predictable residual

For the empirically predictable residual we train a gradient-boosted quantile regressor of each target lead's off-dipole residual on the observed leads and apply Mondrian conformalized quantile regression [8] per (S, s, ℓ) group, with strict fold discipline (basis and model on folds 1–7, calibration on fold 9, a single evaluation on fold 10; nominal $1-\alpha$=0.90). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.88–0.95; e.g. {I,II,V1,V3,V5} QRS/V2 0.92, record-bootstrap CI $[0.89, 0.95]$, mean width 0.165 mV).

What is and is not guaranteed. Mondrian CQR gives *within-(S, s, ℓ)-group marginal* coverage under exchangeability—*not* coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.80 (n=40), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

5 Experiments

Data. PTB-XL [10] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [11]; the map is evaluated on records disjoint from the folds used to estimate $\boldsymbol{\mu}_s, \boldsymbol{M}_s$. All table numbers regenerate from result JSON via `paper/emit_baseline_table.py`.

5.1 The recoverability map on PTB-XL

We estimate $\boldsymbol{M}_s$ (rank 3) per segment on the training folds and evaluate the map on 1,200 NORM records disjoint from the estimation folds (per-lead η, κ with 200-sample bootstrap CIs, truncation

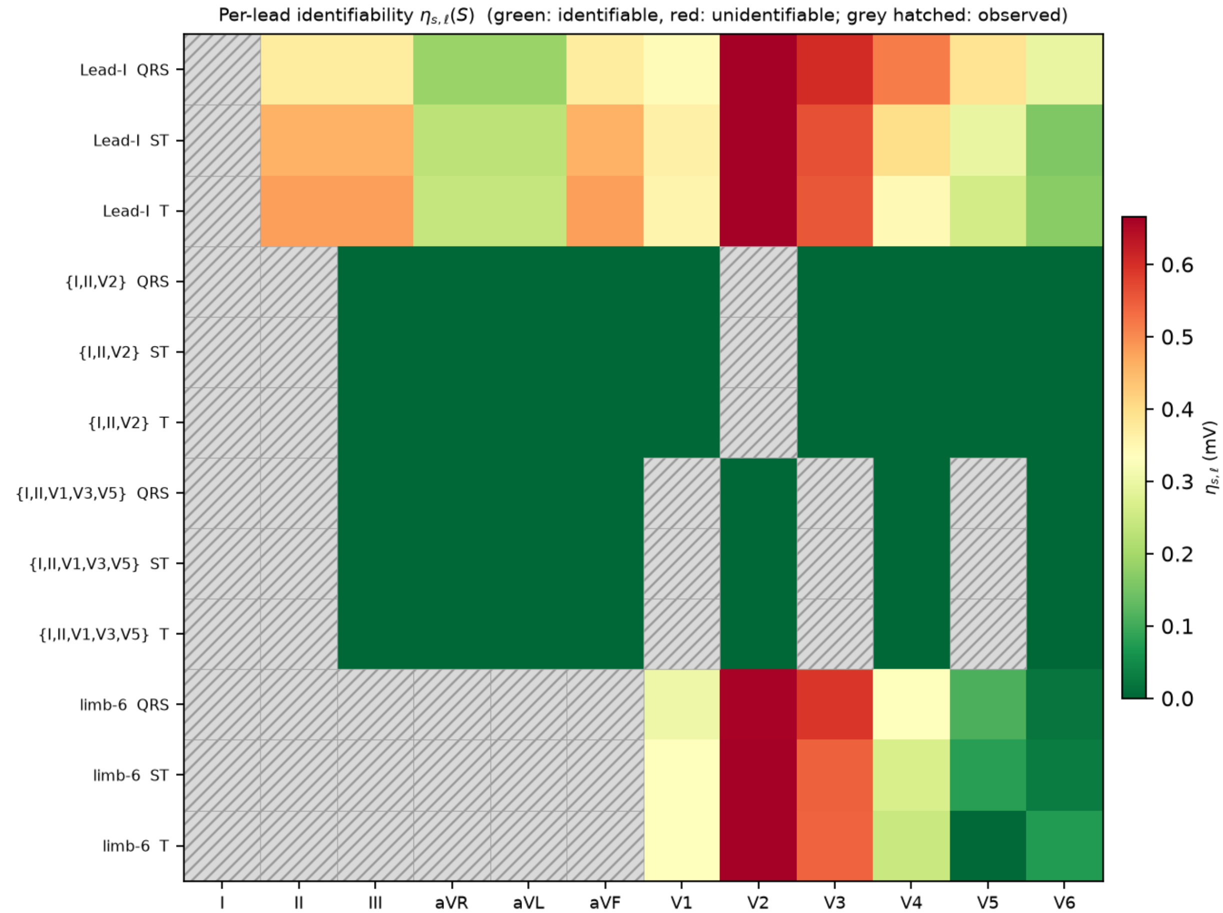

Figure 1: Per-lead identifiability $\eta_{s,\ell}(S)$ across configurations and segments (green: identifiable, red: unidentifiable; grey hatched: observed). For limb-6 the anterior precordial leads are unidentifiable while lateral V5/V6 grade toward identifiable.

tolerance ϱ=10^{-2}; Fig. 1). The map is *graded and per target lead*: a single observed lead (e.g. Lead I) leaves every other lead's dipolar component unidentifiable (η>0; rank 1); a dipole-spanning triplet {I,II,V2} or the five-lead {I,II,V1,V3,V5} makes all targets identifiable ($\eta\approx 0$ to machine precision, since S then spans the rank-3 subspace). The value over a binary rank test is the *gradient* it resolves: for limb-6 the ST identifiability falls smoothly from strongly unidentifiable anterior leads (V2 η=0.712, V3 0.553, V1 0.330, V4 0.268) to *near-identifiable* lateral leads (V5 0.084, V6 0.027). So the honest a priori warning is specific: **anterior precordial ST (V1–V4) is not identifiable from limb leads**—the direction of the estimated subspace it needs is unobserved—while lateral V5/V6 are largely recoverable. A physiological binary "precordial vs. limb" split cannot make this call; the empirical graded map does.

Certificate-level safety is reconstructor-invariant at the floor. The warning is actionable but must be stated on a reconstructor-*invariant* quantity. On 1498 delineated fold-10 records we reconstruct anterior precordial ST from limb-6 with three reconstructors and count ST-threshold events (|ST|>0.1 mV crossings)—false positives (a crossing in the reconstruction, not the truth) and false negatives (vice versa; Table 1). The *total* wrong-event rate is near-invariant across reconstructors (47.1–50.1%, at mean ST error 0.059–0.081 mV): this floor is the cost of the certified

Table 1: Limb-6 → anterior-precordial ST reconstruction, 1498 fold-10 records. FP = false-positive, FN = false-negative ST-threshold crossings (%). The *total* (FP+FN) is near-invariant—the certified floor; the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	ST err (mV)
dipolar	32.5	17.6	50.1	0.076
ridge	5.3	42.3	47.6	0.059
OLS	1.1	46.0	47.1	0.081

Table 2: Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol (800 test records). "dipolar" = Tier I; "ridge" = per-segment ridge (Tier I+II); "U-Net" = strong arbitrary-mask neural baseline. Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error, the gap being the unobserved dipolar coordinate $\eta>0$ flags. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.471	0.234	0.220	**0.181**
{I,II,V1,V3,V5}	0.484	0.186	0.168	**0.164**
limb-6	0.484	0.362	0.346	**0.239**

unidentifiability. What the reconstructor *does* choose is the false-positive/false-negative split—dipolar leans to false positives (32.5%/17.6%), OLS to false negatives (1.1%/46.0%)—a design trade-off, not a certified number. We therefore report the whole matrix and anchor the certificate on the invariant floor, not a single cell.

Truncation-tolerance sensitivity. Rank/κ of well-posed configurations are stable across $\varrho \in \{10^{-4}, \ldots, 10^{-1}\}$ (e.g. {I,II,V1,V3,V5} is rank 3 throughout), while near-rank-deficient sets are sensitive: on synthetic data {V1,V2,V3} is rank 2 at $\varrho=10^{-2}$ but rank 3 with $\kappa\sim 200$ at $\varrho=10^{-4}$. We therefore report a single κ only for well-conditioned configurations and give the ϱ-sweep and bootstrap CIs otherwise; a single 10^4 or 10^5 figure for a degenerate set is not meaningful.

5.2 Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: per-timepoint waveform RMSE on the same 800 held-out fold-10 records, same segment indices, record-bootstrap CIs. We include a *strong neural baseline*: an arbitrary-mask conditional 1-D U-Net trained with random lead masks (so one model serves any configuration, no target-lead leakage), the masked-reconstruction architecture the reduced-lead literature uses [1, 5].

On a dipole-spanning set the methods form a monotone ladder—prior-mean → dipolar (Tier I) → per-segment ridge (Tier I+II) → U-Net—confirming there is recoverable non-dipolar content (the calibrated Tier II layer), yet the strong neural model's margin over a simple ridge is small (QRS 0.164 vs. 0.168 mV on {I,II,V1,V3,V5}). On **limb-6** the U-Net still *lowers* error substantially over the prior mean (QRS 0.484 → 0.239)—it recovers the predictable, largely population-correlated structure—but plateaus at 1.46× its spanning-set error and no reconstructor closes that gap. The residual gap is exactly the *unobserved dipolar coordinate* the map's $\eta>0$ marks as unidentifiable: capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing that coordinate. We therefore scope the certified claim to the dipolar component, not to a blanket "not recoverable."

Empirical, not physical, subspace. On real PTB-XL the estimated $\boldsymbol{M}_s$ shares two directions with the classical inverse-Dower vectorcardiographic space [3,4] (principal angles 2–6°) but its third direction differs materially (max angle 43–55°, bootstrap CIs excluding small angles). We therefore treat $\boldsymbol{M}_s$ as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all certificate quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment.

6 Cross-Dataset Transfer of the Certificate

The certificate is a function of the estimated subspace $\boldsymbol{M}_s$ and the selection S; a fair question is whether it is an artifact of one cohort. We re-estimate $\boldsymbol{M}_s$ and the per-configuration conditioning on the independent Chapman-Shaoxing 12-lead database [12] (∼10,000 patients, different country, devices, and sampling) and compare to PTB-XL (`results/cross_dataset.json`). We deliberately do *not* claim "population independence"; we report what actually transfers.

QRS subspace is transfer-stable. For QRS the two datasets' rank-3 subspaces are nearly aligned (principal angles 20.3°/9.1°/1.6°; explained-variance ratios 0.56/0.27/0.08 vs. 0.55/0.27/0.06). The per-configuration conditioning transfers: κ for {I,II,V2} is 3.1 (PTB-XL) vs. 4.1 (Chapman), for {I,II,V1,V3,V5} 2.5 vs. 2.3, and the limb-6 ill-conditioning is reproduced on both ($\kappa \sim 2 \times 10^5$), i.e. the *qualitative* identifiable/unidentifiable verdict and the *quantitative* κ ordering are stable across cohorts.

ST/T third direction does not transfer. For the low-amplitude segments (P, ST, T) two of three subspace directions align but the third differs sharply between datasets (max principal angle 88–89°). We therefore restrict the transfer claim: the QRS map and the limb-6 conditioning verdict are cohort-stable; the ST/T *third* spatial direction is cohort-specific and its map should be re-estimated per dataset. This is the honest scope of transfer, not a blanket invariance.

7 A Negative Result: the Generative "Fabrication-Objective" Story Did Not Survive

This project began with a stronger claim—that a diffusion reconstructor proves "fabrication is a property of the objective," a certified hallucination. A pre-submission re-analysis retracted it. We document the retraction in full because the negative result is itself informative and because selective reporting of the earlier (favourable) numbers would be exactly the failure the certificate is meant to prevent.

Setup. An arbitrary-mask conditional DDPM (12-channel binary mask + observed values; random-mask training so one model serves any configuration, with explicit leakage tests) reconstructs {V2,V4,V6} from {I,II,V1,V3,V5} under classifier-free-style guidance weight $w \in \{1, 2, 4, 6\}$ (`results/gpu_diffusion_leakfixed.json`, `realism_metrics.json`, `gpu_deficit_ci.json`).

(1) The original effect was a configuration-leakage artifact. The earlier exhibit reused one model trained on a fixed observed set to score a *different* configuration. With leakage removed (arbitrary-mask model + leakage guards) and 5 training seeds, the "achievability gap" $\Delta\rho = \rho_{\text{achievable}} - \rho_{\text{model}}$ is monotone in w but *marginal*: mean $\Delta\rho = -0.031, -0.024, +0.024, +0.058$ for $w = 1, 2, 4, 6$, with ± 0.044 at w=6 ($\approx 1.3\sigma$). The previously reported "4.3σ" does not reproduce.

(2) Realism does not improve with guidance—the causal premise is false. The story required higher guidance to buy realism at the cost of fidelity. Instead both realism metrics *worsen* monotonically with w: the target-lead PSD log-distance rises (V2: 1.95 → 2.35 from w=1 to 6) and the amplitude-Wasserstein distance rises (V2: $0.29 \to 0.67$). High guidance simply degrades the model on every axis; there is no realism/fidelity trade to exploit.

(3) What is true instead. The predictable non-dipolar residual is real but modest: oracle correlations (`gpu_oracle_gate.json`) are $\rho \approx 0.49$–0.55 for V2 (QRS/ST) and 0.2–0.3 for V4/V6—consistent with the calibrated Tier-II intervals of Sec. 4. So the honest statement is a *graded predictability* of the residual, calibrated per feature and lead, not a proof of fabrication. We removed the fabrication-objective claim and rebuilt the paper around the parts that withstand scrutiny.

8 Limitations

The subspace $\boldsymbol{M}_s$ is estimated, so η/κ inherit its estimation error (reported via record-bootstrap CIs) and its cohort dependence (Sec. 6); the ST/T third direction is cohort-specific. Identifiability is stated for the low-rank (dipolar) coordinate: a reconstructor may still recover predictable non-dipolar structure (Sec. 5), so $\eta > 0$ bounds the dipolar component, not every waveform detail. Conformal coverage is within-(S, s, ℓ)-group marginal under exchangeability, not conditional on diagnosis; rare diagnostic subgroups are under-covered (Sec. 4). Delineation uses NeuroKit2 and inherits its errors. The certificate is a pre-reconstruction screen, not a substitute for clinical validation.

9 Conclusion

Reduced-lead ECG reconstruction has a known low-rank structure per segment. Exploiting it gives a graded, per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor, plus a reconstructor-invariant error floor. It converts "trust the reconstruction" into a per-feature, per-lead, calibrated statement, and it told us honestly when one of our own earlier claims was not supported.

A Proof of Proposition 1

Write $\boldsymbol{M}_{s,S} = \mathbf{Sel}_S \boldsymbol{M}_s$ with truncated SVD $\boldsymbol{M}_{s,S} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$ at tolerance ϱ, so $\boldsymbol{M}_{s,S}^+ = \boldsymbol{V}\boldsymbol{\Sigma}^{-1}\boldsymbol{U}^\top$ and $\boldsymbol{P}_{\text{obs}} = \boldsymbol{M}_{s,S}^+ \boldsymbol{M}_{s,S} = \boldsymbol{V}\boldsymbol{V}^\top$ is the orthogonal projector onto $\text{row}(\boldsymbol{M}_{s,S}) = \text{span}(\boldsymbol{V})$.

(i) Identifiability. The dipolar part of lead ℓ is the linear functional $f_\ell(\boldsymbol{d}) = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{d}$. From S we observe $\mathbf{Sel}_S \boldsymbol{M}_s \boldsymbol{d} = \boldsymbol{M}_{s,S}\boldsymbol{d}$, i.e. the coordinates $\boldsymbol{V}^\top \boldsymbol{d}$ (the components of $\boldsymbol{d}$ in $\text{row}(\boldsymbol{M}_{s,S})$) up to noise; the orthogonal complement $(\boldsymbol{I} - \boldsymbol{P}_{\text{obs}})\boldsymbol{d}$ is unobserved. Decompose $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{P}_{\text{obs}} + \boldsymbol{e}_\ell^\top \boldsymbol{M}_s (\boldsymbol{I} - \boldsymbol{P}_{\text{obs}})$. If $\eta_{s,\ell}(S) = \|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s (\boldsymbol{I} - \boldsymbol{P}_{\text{obs}})\|_2 = 0$ then $f_\ell(\boldsymbol{d}) = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{P}_{\text{obs}} \boldsymbol{d}$ depends on $\boldsymbol{d}$ only through the observed coordinates $\boldsymbol{P}_{\text{obs}}\boldsymbol{d}$, hence is determined by $\boldsymbol{M}_{s,S}\boldsymbol{d}$: lead ℓ's dipolar component is identifiable. Conversely if $\eta_{s,\ell}(S) > 0$ there is a unit $\boldsymbol{\delta} \in \ker \boldsymbol{M}_{s,S}$ (an unobserved dipole direction, so $\boldsymbol{M}_{s,S}\boldsymbol{\delta} = 0$) with $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{\delta} \neq 0$; the two dipoles $\boldsymbol{d}$ and $\boldsymbol{d} + t\boldsymbol{\delta}$ produce identical observations for all t but change f_ℓ by $t\, \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{\delta}$, so no estimator recovers f_ℓ at any SNR. This is the estimability criterion [6] for f_ℓ.

(ii) Conditioning. The mean-centred estimate is $\widehat{\boldsymbol{L}}_s - \boldsymbol{\mu}_s = \boldsymbol{M}_s \boldsymbol{M}_{s,S}^+ (\boldsymbol{y}_S - \boldsymbol{\mu}_{s,S})$ with $\boldsymbol{y}_S - \boldsymbol{\mu}_{s,S} = \boldsymbol{M}_{s,S}\boldsymbol{d} + \mathbf{Sel}_S \boldsymbol{r}_s + \boldsymbol{n}$. On the identifiable part its lead-ℓ error from the observed non-dipolar residual and noise is $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{M}_{s,S}^+ (\mathbf{Sel}_S \boldsymbol{r}_s + \boldsymbol{n})$, and by Cauchy–Schwarz $|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{M}_{s,S}^+ (\mathbf{Sel}_S \boldsymbol{r}_s + \boldsymbol{n})| \le \|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{M}_{s,S}^+\|_2 (\|\mathbf{Sel}_S \boldsymbol{r}_s\| + \|\boldsymbol{n}\|) = \kappa_{s,\ell}(S)(\|\mathbf{Sel}_S \boldsymbol{r}_s\| + \|\boldsymbol{n}\|)$. Taking the max over ℓ gives $\kappa_s(S) = \|\boldsymbol{M}_s \boldsymbol{M}_{s,S}^+\|_2$. □

B Reproducibility

Code, fixed seeds, an exact environment lock (`env.lock.txt`), and a one-command pipeline (`experiments/run_all` are released; every table number is regenerated from result JSON (`paper/emit_baseline_table.py`), and a CI workflow runs the test suite on every push. The neural baseline is a 24→12-channel 1-D U-Net (base width 64, residual GroupNorm blocks, 60 epochs, Adam 2×10^{-4}, MSE on masked leads, random-mask training); the residual predictor is a gradient-boosted quantile regressor (pinball loss) with strict fold discipline (folds 1–7 train, 9 calibrate, 10 test).

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